EXPERIMENT No3: 2-way Switching

Student Guide

Prepared by BLIS GLOBAL Robotics Department

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Objective:

In this experiment, you will learn how to wire a light bulb using a 2-way switching circuit, where two switches can control the same light bulb. This wiring method is commonly used in homes where a light can be controlled from two different locations, such as at both ends of a hallway or staircase.

Materials Needed:

- 1 Light Bulb
- 2 SPDT (Single Pole Double Throw) switches
- 1 Battery or Power Supply (appropriate voltage for the light bulb)
- Wires
- Breadboard or mounting board (optional)

What is 2-Way Switching?

A 2-way switch allows you to control a light from two different locations. The circuit uses two SPDT switches, which can direct the current flow in different paths. Depending on the position of each switch, the light bulb will either be powered on or off.

Theory Behind 2-Way Switching:

SPDT Switch (Single Pole Double Throw): This switch has three terminals:

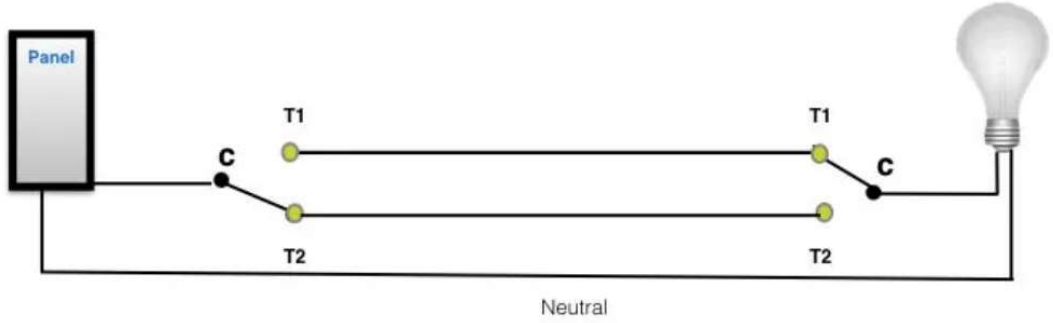
- Common (C): The terminal where the power is connected.
- L1 and L2: These terminals connect to the second switch and the bulb.

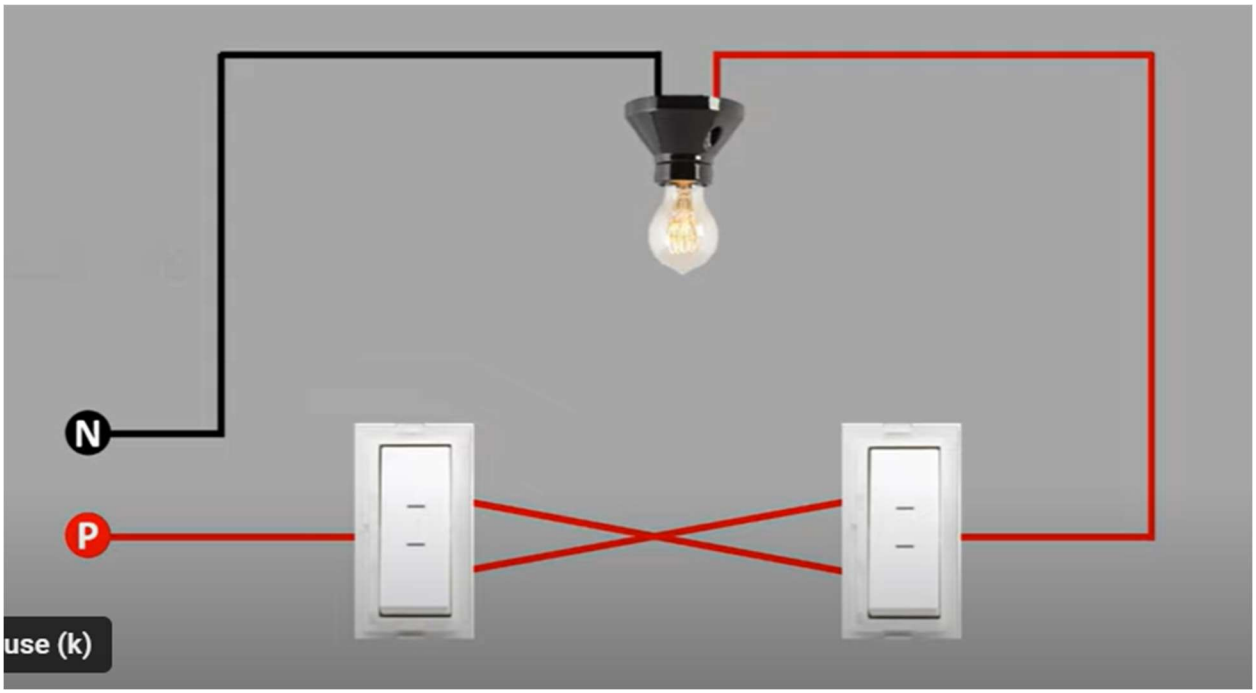
In a 2-way switch setup, each switch alternates the path of current, allowing the light to be turned on or off from either switch.

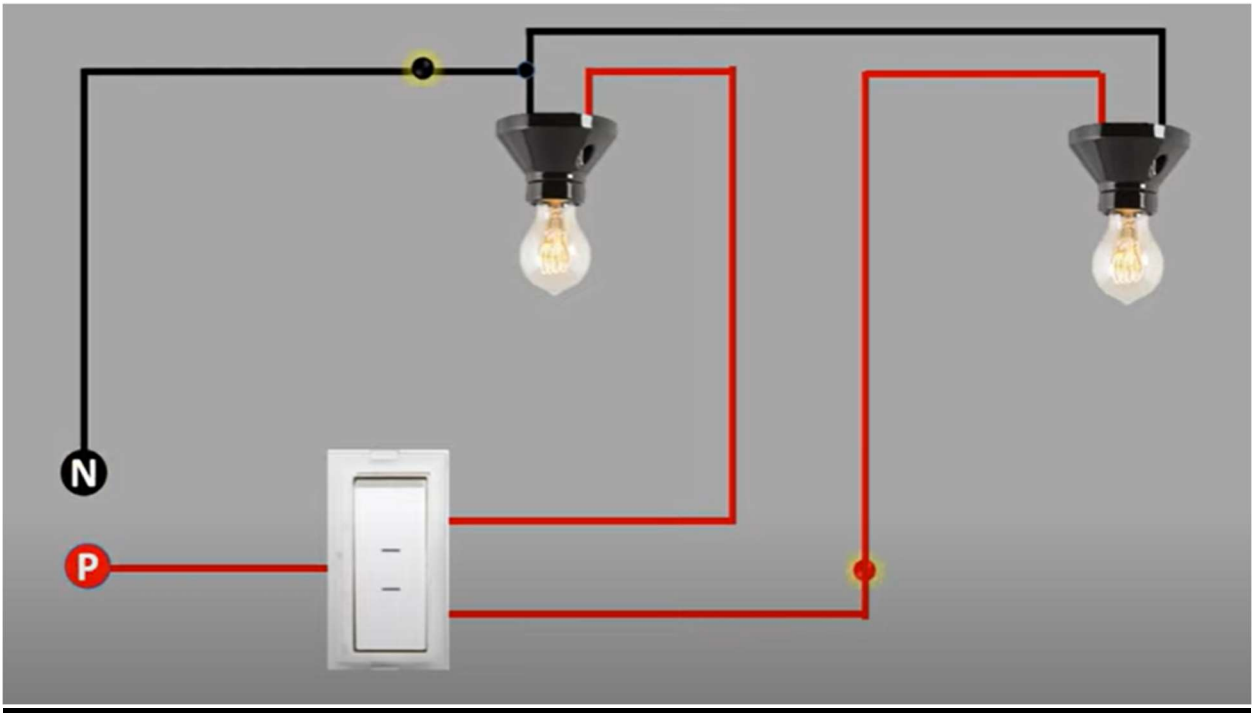
Steps to Set Up the Circuit:

1. **Connect the First Switch:**
   - Connect the common terminal (C) of the first switch to the positive terminal of the battery or power source.
   - Connect the L1 terminal of the first switch to the L1 terminal of the second switch.
   - Connect the L2 terminal of the first switch to the L2 terminal of the second switch.
2. **Connect the Second Switch:**
   - Connect the common terminal (C) of the second switch to one terminal of the light bulb.
3. **Complete the Circuit:**
   - Connect the other terminal of the light bulb to the negative terminal of the battery or power source.
4. **Test the Circuit:**
   - Flip either switch. You will notice that the light turns on or off depending on the switch positions.
   - Flip the second switch to see how it also controls the light from the second location.

Circuit Diagram







Wiring Diagram Overview:

- Switch 1 (SPDT) connects to the power source.
- Switch 2 (SPDT) connects to the light bulb.
- L1 and L2 terminals on both switches are cross-connected.

This arrangement allows the current to flow through the light bulb when both switches are aligned in the correct positions. If the positions of either switch are changed, the current flow is interrupted, and the light turns off.

Observations:

- What happens when you flip one switch? Does the light turn on or off?
- How does flipping the other switch change the state of the light?
- Can you turn the light on from one switch and off from the other?

Theory Recap:

- In a 2-way switch circuit, two switches are connected in such a way that either switch can control the light.
- This circuit uses two SPDT switches, and the position of either switch determines whether the circuit is closed (light on) or open (light off).

Applications of 2-Way Switching:

- Staircases: Control a light at the top and bottom of the stairs.
- Hallways: Control a light from either end of a hallway.
- Rooms with multiple entrances: Control a light from different entry points.

Conclusion:

In this experiment, you learned how to wire a light bulb using 2-way switching. This type of wiring is useful in real-world applications where controlling a light from multiple locations is required. You now understand how SPDT switches can be used in tandem to create a versatile and practical lighting circuit.

Challenge:

- Try adding another switch (3-way switch system) to see if you can control the light from three locations!